

Phase 4: racing together

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Phase 3 is on the server and finished. Phase 4 turns one driver at a time into a real race, and adds the four things you asked for after testing.

Part	What it is
Races you make and join	Public or invite only, with a Start Race button.
One grid, one result	Everybody lined up together, finishing order, gaps.
The pit checkpoint	Fuel to full and repair with no time added.
Speed zones	A limit between any two checkpoints you pick.
The spare tire	Put the air back without touching the rest of the car.

1 Races you make and join

Today a run arms the moment you pick a course, and your time starts the first time you touch the start line. That is what you called touchy, and you are right. It is easy to set off by accident.

A race becomes something you make, and other people join.

1. You pick a course, the laps and the mode, and you make a race.
2. You choose **public**, so anyone on the server can join, or **invite**, and you pick who from the player list.
3. People join and are put on the grid behind the line.
4. Nothing is armed. The line does nothing. Sit there as long as you like.
5. You press **Start Race**. Only then does the line go live.
6. Each driver's time starts when they cross the first checkpoint.

The line still starts your own time, which is what keeps the timing fair when people cross it a second apart. It just does nothing at all until you say go.

```
local lobbies = {}

function RM.lobby.create(pid, d)
  local track = RM.tracks.get(tostring(d.track or ""))
  if not track then return false, "no_such_track" end
  if RM.lobby.of(pid) then return false, "already_in_one" end

  local id = RM.util.shortId()
  lobbies[id] = {
    id      = id,
    host    = pid,
    track   = track.id,
    laps    = math.floor(tonumber(d.laps) or 1),
    mode    = tostring(d.mode or "controller"),
    open    = d.open and true or false,
    state   = "waiting",
    members = { [pid] = true },
    invited = {},
  }
  return true, lobbies[id]
end

function RM.lobby.join(pid, id)
  local l = lobbies[tostring(id or "")]
  if not l then return false, "no_such_race" end
  if l.state ~= "waiting" then return false, "already_started" end
```

```

    if RM.lobby.of(pid) then return false, "already_in_one" end

    local key = RM.identity.keyOf(pid)
    if not l.open and not l.invited[key] then return false, "not_invited" end

    l.members[pid] = true
    return true, l
end

function RM.lobby.start(pid)
    local l = RM.lobby.of(pid)
    if not l then return false, "no_lobby" end
    if l.host ~= pid then return false, "not_the_host" end
    if l.state ~= "waiting" then return false, "already_started" end

    l.state = "armed"
    for member in pairs(l.members) do
        RM.race.arm(member, { id = l.track, mode = l.mode, laps = l.laps })
        RM.bus.queue(member, "race.state", RM.race.wire(member))
    end
    return true, l
end

```

2 One grid, one result

Everyone in the race is placed in a row behind the start line rather than all on the same spot. The results come out as one table in finishing order, with the gap to the leader and the gap to the car in front, which is already how the results screen is built.

Somebody who leaves half way no longer holds up the people still driving. They are filed as a did not finish and the race carries on.

3 The pit checkpoint

A checkpoint can be marked as a pit when you build the course. It is drawn in a different colour so it is obvious on the map.

A pit is not part of the lap. Driving through it does not count as a gate and does not skip one. What it does is put you in the pit while you are inside it.

While you are in the pit:

- Fuel fills to 100% instead of adding a quarter.
- Repair and Spare tire add nothing to your race time.
- The holds still apply in full, and the clock is still running.

So a pit stop costs you the time it takes and nothing on top. That is the whole point of having one.

```

function RM.service.use(pid, d)
    local action = ACTIONS[tostring(d.which or "")]
    if not action then return false, "no_such_action" end
    if jobs[pid] then return false, "already_working" end

    local r = RM.race.get(pid)
    local racing = r ~= nil and r.state == "running"
    local inPit = racing and r.inPit == true

    local hold, cost = 0, nil
    if racing then
        hold = tonumber(RM.config.holds[action.hold]) or 0
        if action.penalty and not inPit then
            local seconds = tonumber(RM.config.penalties[action.penalty])
            if seconds and seconds > 0 then

```

```

        RM.race.penalty(pid, seconds, action.penalty)
        cost = seconds
    end
    end
end

jobs[pid] = { which = action.name, hold = hold, endsAt = RM.now() + hold,
              cost = cost, full = inPit }
return true, jobs[pid]
end

```

4 Speed zones between checkpoints

You asked whether the 37 mph limit could go between chosen waypoints. Yes, and it is one of the easier things in here.

The checkpoints are already numbered, so a zone is just a start, an end and a speed. A course can have as many as you like, and they can overlap a pit or not, as you prefer.

```

zones = {
  { from = 12, to = 14, mph = 37 },
}

```

The server already reads everybody's speed for the player list, so it does the checking. Nothing is trusted to the driver's own game.

Going over the limit does not punish you instantly. You get a warning on screen, and the penalty only lands if you stay over it, so one bump over a crest costs nothing. Once it has charged you it will not charge again until you drop back under.

```

function RM.zones.sample(pid, mph)
  local r = RM.race.get(pid)
  if not r or r.state ~= "running" then return end

  local zone = RM.zones.at(r.track, r.nextGate)
  if not zone then r.overFor, r.overCharged = nil, nil return end

  if mph <= zone.mph then
    r.overFor, r.overCharged = 0, false
    return
  end

  r.overFor = (r.overFor or 0) + (RM.config.rosterMs / 1000)
  if r.overFor >= RM.config.speedGrace and not r.overCharged then
    r.overCharged = true
    RM.race.penalty(pid, RM.config.penalties.speeding, "speeding", r.nextGate)
  end
end

```

5 The spare tire

I told you this could not be done. I looked again after you said you had seen it done somewhere, and I was wrong. There is a way.

A flat in this game is two separate things. The air is gone, and some of the tire is torn. There is no call that mends a tire, which is what I found the first time and why I said no. But the engine can put air back into a tire, and the game already does exactly that for the trucks with onboard air compressors. Every wheel also remembers the pressure it started at.

So the plan is to put the air back and clear the flat, in place, and leave the rest of the car alone.

```

for _, wd in pairs(wheels.wheels) do

```

```

if wd.isTireDeflated and wd.pressureGroupId and wd.startingPressure then
  obj:setGroupPressure(wd.pressureGroupId, wd.startingPressure)
  wd.isTireDeflated = false
end
end
end

```

The honest part. This will work for a tire that went flat from a puncture, which is nearly every flat you get in a race. If a tire is properly shredded off the rim the game deflates the whole group and changes how it drags, and that will not come back from air alone. In that case the button falls back to what it does now, which is the full repair, and it will tell you why.

Because it needs trying in the game and might only cover the common case, this one sits on its own rather than inside the main phase 4 work.

The tire shop sound goes on the same button. Send me a file if you have one you like, otherwise I will use one of the game's.

6 The messages

New channels, in the envelope everything already uses.

	Channel	Carries
up	race.create	course, laps, mode, public or invite
up	race.join	the race id
up	race.invite	who to invite
up	race.leave	nothing
up	race.start	nothing, and only the host may send it
down	race.lobbies	the open races, for the list
down	race.lobby	the one you are in, and who else is in it
down	pit.state	you are in the pit, or you are not
down	zone.warn	you are over the limit, and by how much

7 Order of work

1. The pit checkpoint. Small, self contained, useful on its own.
2. Speed zones. Also small, and it uses the pit work.
3. Lobbies on the server: make, join, invite, start. All of it testable without opening the game.
4. The race panel: open races, a join button, an invite list, Start Race.
5. One grid for everybody and one set of results.
6. The spare tire experiment, last, because it is the only part I cannot promise the shape of before trying it.

Server first every time. Most of it is proven before the game is opened, which is what worked in phases 2 and 3.

8 Testing

- Mock tests for the lobby: only the host can start, an invite only race refuses a stranger, a race that has started refuses a late joiner, and a host who leaves hands the race to somebody else.
- Mock tests for the pit: holds still apply, penalties do not, fuel goes to full.

- Mock tests for zones: under the limit costs nothing, a brief bump over costs nothing, staying over charges once and not twice.
- In game with two cars for the grid and the results.
- In game for the tire, which is the only part that cannot be proven any other way.

9 Questions, and what I will do if you say nothing

	Question	Default if you say nothing
1	How long over the limit before it charges you?	Three seconds. A bump over a crest costs nothing.
2	How much is a speeding penalty?	Thirty seconds, the same as a missed checkpoint.
3	Can people join after Start Race?	No. Pressing it closes the field.
4	What if the host leaves before starting?	The race passes to whoever joined first.
5	More than one race at once?	Yes, on different courses. One race per course.
6	Who can make a public race?	Anybody.
7	If a tire is too far gone for air alone, what should the button do?	Fall back to the full repair and say so.
8	Should the pit fix a flat for free as well?	Yes. In the pit nothing adds to your time.

10 Not in phase 4

Records and personal bests are phase 5. CoPilot and Challenges are phase 6.

The look of the interface is a separate job. It can happen whenever you want and does not need to wait for phase 6.

Phase 3 is on the server now. Phase 4 starts on your word, and the eight questions above are the only thing that would change the shape of it.